

# Seunghyun An

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*Research interests: Operating systems, chiplet-based server systems, and performance engineering.*

## EDUCATION

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### University of Wisconsin–Madison

Madison, WI, USA

*Ph.D. in Computer Science*

*Aug. 2024 – Present*

- GPA: 4.0/4.0
- Doctoral Minor in Astronomy

### Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

*B.S. in Computer Science and Electrical Engineering (Double Major)*

*Mar. 2018 – Aug. 2024*

- GPA: 3.88/4.3, Magna Cum Laude
- Honor Student

## PUBLICATIONS

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### cSwitch: Scheduling Linux Threads under the I/O Chiplet Wall

SOSP '26

*Seunghyun An, Joontaek Oh, Ming Liu* [code](#)

- Showed that I/O-chiplet contention creates placement and load-balancing pathologies invisible to CPU-centric Linux schedulers.
- Designed and implemented cSwitch, a traffic-aware Linux scheduling framework that reduced latency, increased throughput, and improved scalability over existing schedulers with modest overhead.

### Server Chiplet Networking

HotNets '25

*Seunghyun An, Joontaek Oh, Ming Liu* [paper](#)

- Characterized communication across compute/I/O chiplets, memory, and devices on two AMD EPYC generations.
- Identified topology-dependent latency and bandwidth domains, inconsistent bandwidth-delay products, and sender-driven sharing.

## RESEARCH EXPERIENCE

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### Operating Systems for Chiplet Servers

Madison, WI

*NetLab, UW-Madison (Advisor: Prof. Ming Liu)*

*Aug. 2024 – Present*

- Designed controlled memory/CXL experiments and hardware-counter analysis to isolate compute-I/O chiplet contention from downstream memory-controller saturation on AMD EPYC systems.
- Implemented cSwitch's Linux `sched_ext`/eBPF path and PMU-based per-thread load attribution.
- Developed bandwidth-aware thread migration and CPU-time throttling policies.
- Investigating selective message passing for coherence-heavy inter-chiplet coordination in microkernel and multikernel designs.

### Kernel-Level RPC for SmartSSD

Daejeon, South Korea

*CASYS, KAIST (Advisor: Prof. Youngjin Kwon)*

*Jul. 2023 – Feb. 2024*

- Ported a user-space RDMA RPC library into a Linux kernel module for SmartSSD services.
- Adapted memory management and synchronization to kernel constraints.

### Concurrent Systems and File Systems in Rust

Lausanne, Switzerland

*RS3Lab, EPFL (Advisor: Prof. Sanidhya Kashyap)*

*Jan. 2023 – Feb. 2024*

- *MV-RLU*: Ported the protocol to Rust, preserving multi-version synchronization while confining raw-pointer operations to a small `unsafe` core.
- *ArckFS* (Trio, SOSP '23): Ported core paths from C to Rust and built an `mmap`-backed mock kernel interface.

### CofferOS Physical Memory Management

Daejeon, South Korea

*CASYS, KAIST (Advisor: Prof. Youngjin Kwon)*

*Jul. 2022 – Dec. 2022*

- In CofferOS (later published at [EuroSys '26](#)), traced deallocation overhead to page zeroing and buddy coalescing.
- Implemented a deferred-free queue and background kernel task to move page reclamation off the deallocation path.

HONORS

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|                                                                     |                                    |
|---------------------------------------------------------------------|------------------------------------|
| Presidential Science Scholarship                                    | 2018 – 2024                        |
| Dean’s List, School of Engineering                                  | Spring 2021                        |
| Gold Medal and First Overall, 22nd International Astronomy Olympiad | Group Beta (senior division), 2017 |

TECHNICAL SKILLS

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|------------------------|---------------------------------------|
| Programming Languages: | C, C++, Rust, Python                  |
| Spoken Languages:      | Korean (native), English (proficient) |

TEACHING EXPERIENCE

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|----------------------------------------------|---------------------------------|
| Teaching Assistant at UW–Madison             |                                 |
| CS 354: Machine Organization and Programming | Spring 2026/Fall 2026 (Head TA) |
| CS 537: Introduction to Operating Systems    | Fall 2024; Spring/Fall 2025     |
| At KAIST                                     |                                 |
| Humanity/Leadership III: Hacking Class       | Fall 2023                       |
| Python Tutor, Daeduk High School             | 2022                            |
| Freshman Tutor: Introduction to Programming  | Spring 2022                     |
| Freshman Tutor: General Physics II           | Fall 2021                       |

PROFESSIONAL EXPERIENCE

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|----------------------------------------------|-------------------------|
| Military Service                             | Pyeongtaek, South Korea |
| KATUSA, Eighth Army Public Affairs           | 2019 – 2020             |
| • Received the U.S. Army Commendation Medal. |                         |